# EC-341 Digital Systems Design
# Experiment # 9

Implementation of Combinational Circuits using EDA Playground

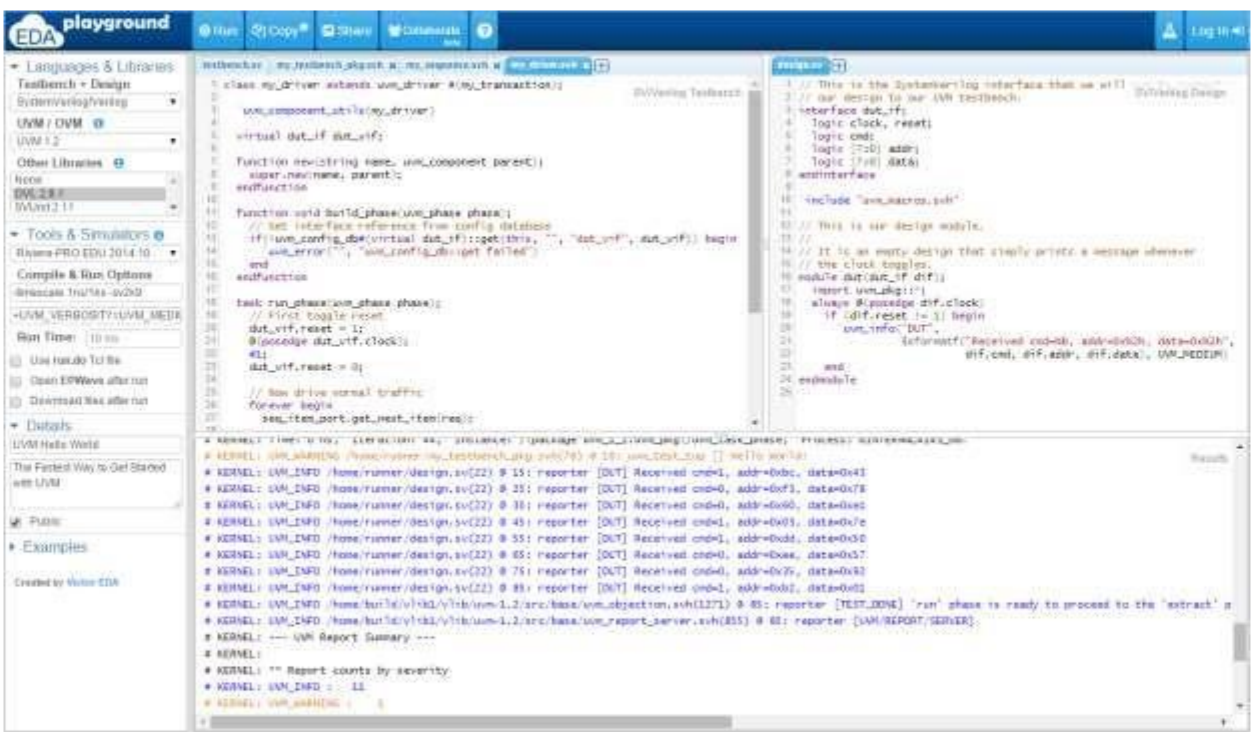
(Verilog Online Simulator)

## Objectives:

- Overview of EDA Playground Environment
- Example
- Task

## 1. EDA Playground

EDA Playground gives us an immediate hands-on exposure to simulating System Verilog, Verilog, VHDL, C++/System C, and other HDLs online. All we need is a web browser. With a simple click, we can run our code and see console output in real time. We can also view waves for simulation using EPWave browser-based wave viewer.



So, as you can see, there are is a left panel of the windows and two workspaces. The left panel offers all the options for selecting between different Programming Languages, Tools and Simulator and Examples which can be directly run for different languages. Some of the main feature that the left panel offers are:

## 1.1 Tools & Simulators

Available tools and simulators are below. EDA Playground can support many different tools.

### 1.1.1 Simulators

- **Synopsys VCS**
  Commercial simulator for VHDL and SystemVerilog
- **Cadence Incisive**
  Commercial simulator for VHDL and SystemVerilog (VHDL simulation not yet implemented on EDA Playground)
- **Aldec Riviera-PRO**
  Commercial simulator for VHDL and SystemVerilog
  Riviera-PRO Product Manual (registration required)
- **Incisive Specman Elite**
  Commercial simulator that supports e Verification Language, IEEE 1647
  Works with Cadence Incisive
  Hello e World Video Tutorial
- **GHDL**
  an open-source simulator for the VHDL language
  fully supports the 1987, 1993, 2002 versions of the IEEE 1076 VHDL standard and partially the latest 2008 revision
- **Icarus Verilog**
  Version 0.10.0 (devel) supports several System Verilog features.
- **GPL Cver**
- **VeriWell**

## 1.2 Compilers and Interpreters

- C++
- Perl
- Python
- Csh (C Shell)

## 1.3 Synthesis Tools

NOTE: The synthesis tools will only process code in the right *Design* pane. The code in the left *Testbench* pane will be ignored.

- Yosys
- Yosys on GitHub
- The Verilog-to-Routing (VTR) Project

### 1.4 Frameworks

Available frameworks:

- SystemVerilog and Verilog
- Python

### 1.5 Libraries & Methodologies

Available libraries and methodologies:

- SystemVerilog and Verilog
- VHDL
- C++

## 2. Example

In the video tutorial, an example is run on the simulator in which an adder's test bench is simulated and resulting waveform in explained as well. Refer to the Video Tutorial for further understanding of the tool.

# 3. Tasks

1. Write a Gate Level Verilog Code for 1bit Half Adder and Full Adder. Verify Results using different instances on test bench.

2. Write a Gate Level Verilog Code for 4bit Adder by using Half and Full Adders. Verify Results for different instances on test bench.

3. Write a Gate Level Verilog Code for 8bit Adder by using 4bit Adders. Verify Results for different instances on test bench.